

Program 1:

```
#include <stdio.h>
int main(){
printf("Welcome to operating system\n");
return 0;
}|
```

Output:

```
crazyguy64@CrazyGuy64:~$ cd OS
crazyguy64@CrazyGuy64:~/OS$ nano hello.c
crazyguy64@CrazyGuy64:~/OS$ gcc hello.c -o hello
crazyguy64@CrazyGuy64:~/OS$ ./hello
Welcome to operating system
```

Program 2:

```
#include <stdio.h>
int main(){
char name[30];
printf("Enter your name\n");
scanf("%s", name);
printf("Welcome %s\n", name);
return 0;
}
```

Output:

```
crazyguy64@CrazyGuy64:~/OS$ nano program2.c
crazyguy64@CrazyGuy64:~/OS$ gcc practice2.c -o practice2
cc1: fatal error: practice2.c: No such file or directory
compilation terminated.
crazyguy64@CrazyGuy64:~/OS$ gcc program2.c -o program2
crazyguy64@CrazyGuy64:~/OS$ ./program2
Enter your name
Dhruv
Welcome Dhruv
```

Program 3:

```
#include <stdio.h>
#include <unistd.h>
int main(){
printf("Process ID =%d\n", getpid());
return 0;
}|
```

Output:

```
crazyguy64@CrazyGuy64:~/OS$ nano program3.c
crazyguy64@CrazyGuy64:~/OS$ gcc program3.c -o program3
crazyguy64@CrazyGuy64:~/OS$ ./program3
Process ID =1296
```

Program 4:

```
#include <stdio.h>
#include <unistd.h>
int main(){
fork();
printf("Hello\n");
return 0;
}
```

Output:

```
crazyguy64@CrazyGuy64:~/OS$ nano program4.c
crazyguy64@CrazyGuy64:~/OS$ gcc program4.c -o program4
crazyguy64@CrazyGuy64:~/OS$ ./program4
Hello
Hello
```

Program 5:

```
#include <stdio.h>
#include <unistd.h>
int main(){
    pid_t pid;
    pid = fork();
    if(pid == 0){
        printf("I am Child Process\n");
    }else{
        printf("I am Parent Process\n");
    }
    return 0;
}
```

Output:

```
crazyguy64@CrazyGuy64:~/OS$ nano program5.c
crazyguy64@CrazyGuy64:~/OS$ gcc program5.c -o program5
crazyguy64@CrazyGuy64:~/OS$ ./program5
I am Parent Process
I am Child Process
```

Program 6:

```
#include <stdio.h>
#include <unistd.h>
int main(){
    fork();
    printf("Hello\n");
    fork();
    printf("hello\n");
    return 0;
}
```

Output:

```
crazyguy64@CrazyGuy64:~/OS$ nano program6.c
crazyguy64@CrazyGuy64:~/OS$ gcc program6.c -o program6
crazyguy64@CrazyGuy64:~/OS$ ./program6
Hello
Hello
hello
hello
hello
hello
```

Program 7:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main(){
    pid_t pid;
    pid = fork();
    if(pid==0){
        printf("Child Running\n");
    }else{
        wait(NULL);
        printf("Parent Running After Children\n");
    }
    return 0;
}
```

Output:

```
crazyguy64@CrazyGuy64:~/OS$ nano proqram7.c
crazyguy64@CrazyGuy64:~/OS$ gcc proqram7.c -o proqram7
crazyguy64@CrazyGuy64:~/OS$ ./proqram7
Child Running
Parent Running After Children
```

Program 8:

```
#include<stdio.h>
#include<unistd.h>
int main(){
    pid_t pid;
    pid=fork();
    if(pid == 0){
        printf("Child pid=%d\n", getpid());
        printf("Parent pid=%d\n", getpid());
    }else{
        printf("Parent pid=%d\n", getpid());
    }
    return 0;
}
```

Output:

```
crazyguy64@CrazyGuy64:~/OS$ nano prohran7.c
crazyguy64@CrazyGuy64:~/OS$ nano program8.c
crazyguy64@CrazyGuy64:~/OS$ gcc program8.c -o program8
crazyguy64@CrazyGuy64:~/OS$ ./program8
Parent pid=1822
Child pid=1823
Parent pid=1823
```